

Problem Statement

Accelerating RAG

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In Retrieval-Augmented Generation (RAG) systems a lot of searching across large knowledge databases occurs to obtain a sense of context before generating a response, However the retrieval stage is extremely memory intensive and suffers from high data movement between the memory and its computing units which leads to a major performance bottleneck, While The CPU and GPU Architectures struggle to maintain computations efficiently on a frequent basis it also leads to a latency and energy overheads, this design aims to to manipulate hardware acceleration techniques and to create a near memory accelerator capable of performing multiple operations directly within the memory module in order to reduce data movement and overall boost the performance of the RAG system.

In addition to the computational bottleneck, the retrieval process in modern RAG models also requires handling large embedding vectors, frequent similarity computations, and Approximate Nearest Neighbor (ANN) searches, which further increase the system's memory bandwidth demand. The accelerator will therefore focus on embedding search optimization, vector quantization, and in-memory dot product operations, enabling faster retrieval without transferring massive datasets between processing units.

The main difficulty lies in the retrieval process's speed-quality trade off where if you wish to boost an aspect some source of sacrifice must occur, in order to assure acceptable quality with noticeable speedup in retrieval time we need to accelerate smarter retrieval algorithms, creating an opportunity to co-design that adapts algorithms and hardware together for the best RAG performance possible.

Furthermore, this project will explore **hardware–software co-design** principles that integrate near-memory computing units with optimized retrieval models, aiming to achieve a balance between **low-latency response**, **high retrieval precision**, and **energy efficiency**. The outcome is expected to demonstrate that **co-optimized near-memory acceleration** can significantly reduce retrieval latency while maintaining high-quality contextual responses, establishing a scalable solution for next-generation RAG systems.